

religion_project

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```
library(readr)
library(dplyr)
library(tidyverse)
# import data
data <- read_csv("Test.csv")
```

This dataset has information about over 36,000 individuals, with columns including information about the state or origin, happiness, questions about attitudes about other religions and immigrants, religious teachings, perceived spirituality/religiosity, marriage status, political leaning, family income, among others. This data comes from the Pew Research Center, a reputable source for surveys on the American population.

Ideas and Questions:

Some potential areas we could explore with this data include seeing which variables are most associated with religiosity.

- 1) Test people's faithfulness as predictors or stand-ins. How important is religion in your life? How important is this to you when making decisions about right and wrong?
- 2) Are you personally a member of a church, synagogue, mosque, temple, or other house of worship?
- 3) Look at people who consider themselves religious vs actually attend religious services.
- 4) Try to find the correlation between consider yourself religious and attendance. Could also use a composite for praying and attendance to as a stand-in for "religiosity"
- 5) Check how the religion related with family or parents. How parents religion affect your religion? Is father religion or mother religion same as your?
- 6) How religious based of age(birth)? Try to find out did the state affect how you religious?
- 7) How marriage affect the religion?
- 8) How support for abortion vs gay marriage as a predictor for party affiliation?

Introduction

The dataset that we're working with comes from the PEW Research Center. The survey was intended to get responses from as many Americans as possible, offering surveys in English and Spanish in all 50 states, with online, paper and phone format options. The data contains information about the religious, political, spiritual and social background of those surveyed. The data can be accessed at the following link:

<https://www.pewresearch.org/dataset/2023-24-religious-landscape-study-rls-dataset/> After trimming down our dataset to include only those with useful information and those without all NA values, we were left with 93 columns. Our driving question is twofold: 1) How well can we predict religiosity based on factors including church attendance, family relationships, marriage, political opinions. 2) Which factors are most effective in predicting religiosity of individuals. Does this vary by religious affiliation.

Exploratory Data Analysis

In our dataset, the most important variables are: RELTRAD (religious affiliation) CURREL (same as before but collapses evangelical, mainline and black prot into one group) RELPER (how religious the respondent is, 1-4) SPIRPER (how spiritual the respondent is, 1-4) ATTNDPERRLS (how often they attend in person) GOD (belief in God) HVN (is there a heaven) HLL (is there a hell) PRAY (frequency of prayer in personal life) PRAC_B (how often do you read scripture outside of worship service) PRAC_D (how often do you share your beliefs with others)

First, we turn CURREL into a factor so that we can create “bins” of how many people fall into each religious tradition

```
data <- data |> mutate(CURREL = factor(CURREL, levels=c(1000,10000,20000,30000,40001,40002,50000,60000,70000),
labels=c("Protestant", "Catholic", "Mormon", "Orthodox", "JW", "Other Faith")))
```

From here, we can look at the relationship between each of the other variables and religion.

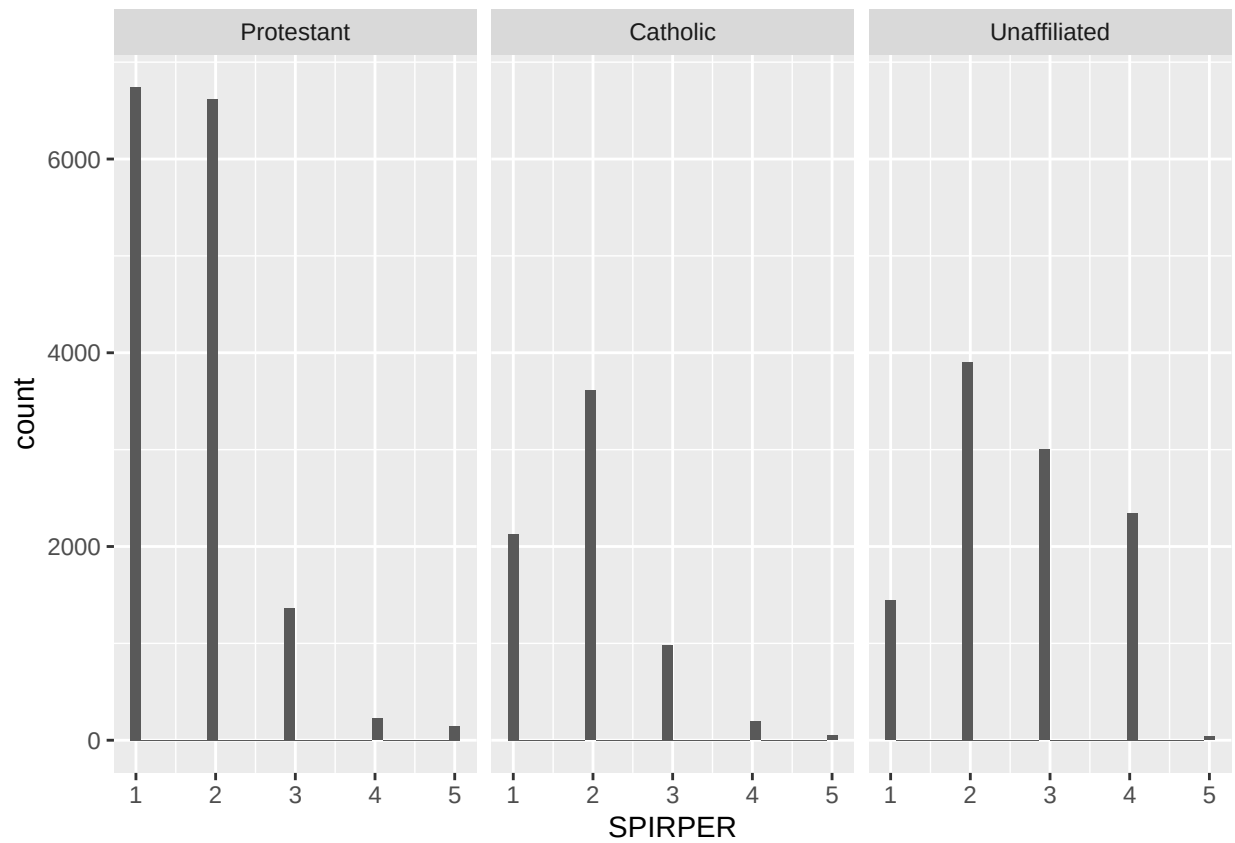
First, before trying to plot the data, we noticed that there was a much larger number of those who identified as Catholic, Protestant or Unaffiliated than anything else, so we split the histogram facet_wrap into two to account for this (then split it again to account for the much larger prevalence of Mormon, Jewish and Other Faith than all the other religious identities combined) so that the spread of each religion across the different values would be clear. This splitting up of the religious categories will prove useful throughout the duration of the EDA section.

Looking at the correlation between each specific faith and personal spirituality (1=most spiritual, 4= least spiritual, 5=don't know/refused), we notice a few trends: most faiths seemed to have the largest spike at either 1 or 2. Interestingly, the Unaffiliated category also has the largest spikes around 1 and 2, and not 4. This could be that many people who identified as Unaffiliated, although they did not profess a specific religion, still considered themselves spiritual in some sense.

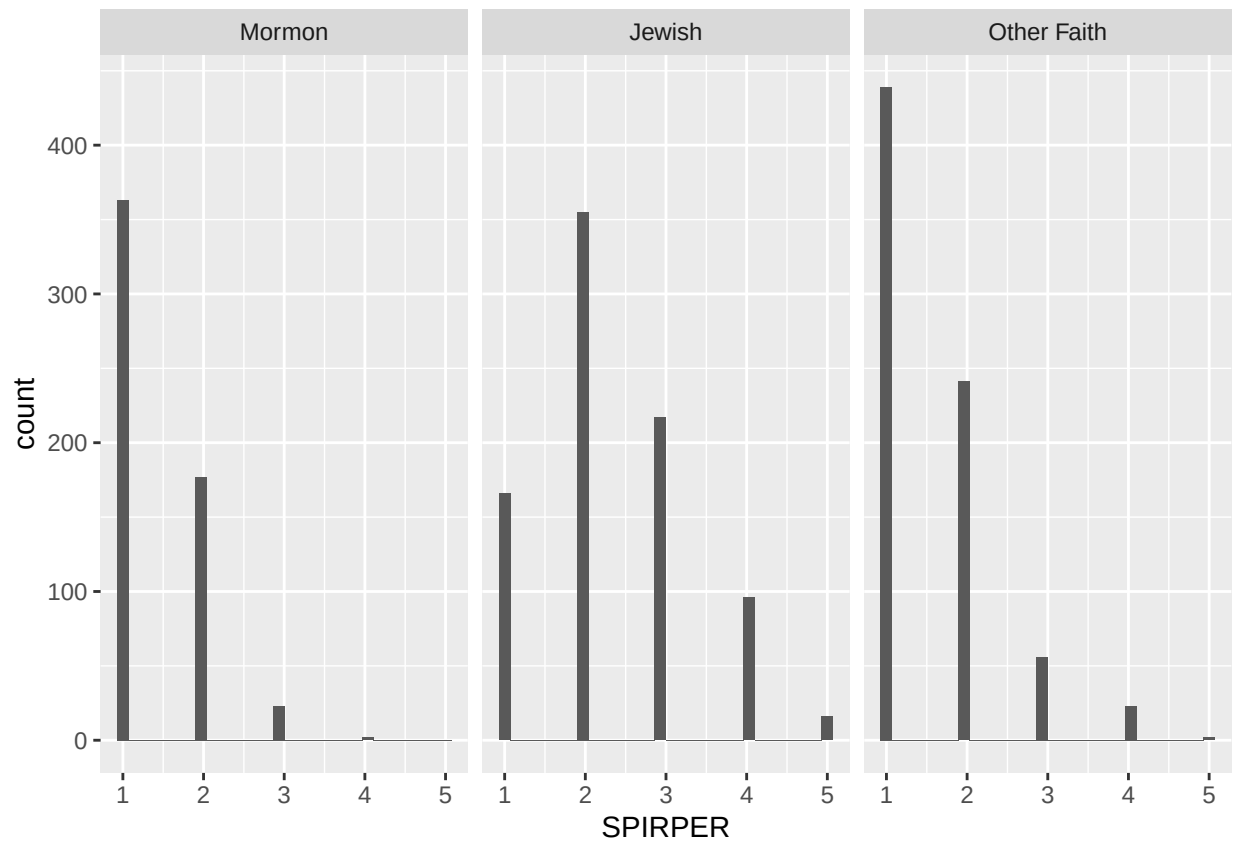
```
data$SPIRPER[data$SPIRPER == 99] <- 5

data_subset1 <- data |>
  filter(CURREL %in% c("Protestant", "Catholic", "Unaffiliated"))
data_subset2 <- data |>
  filter(CURREL %in% c("Mormon", "Jewish", "Other Faith"))
data_subset3 <- data |>
  filter(CURREL %in% c("Orthodox", "JW", "Other Christian", "Muslim", "Buddhist", "Hindu", "Other Relig"))

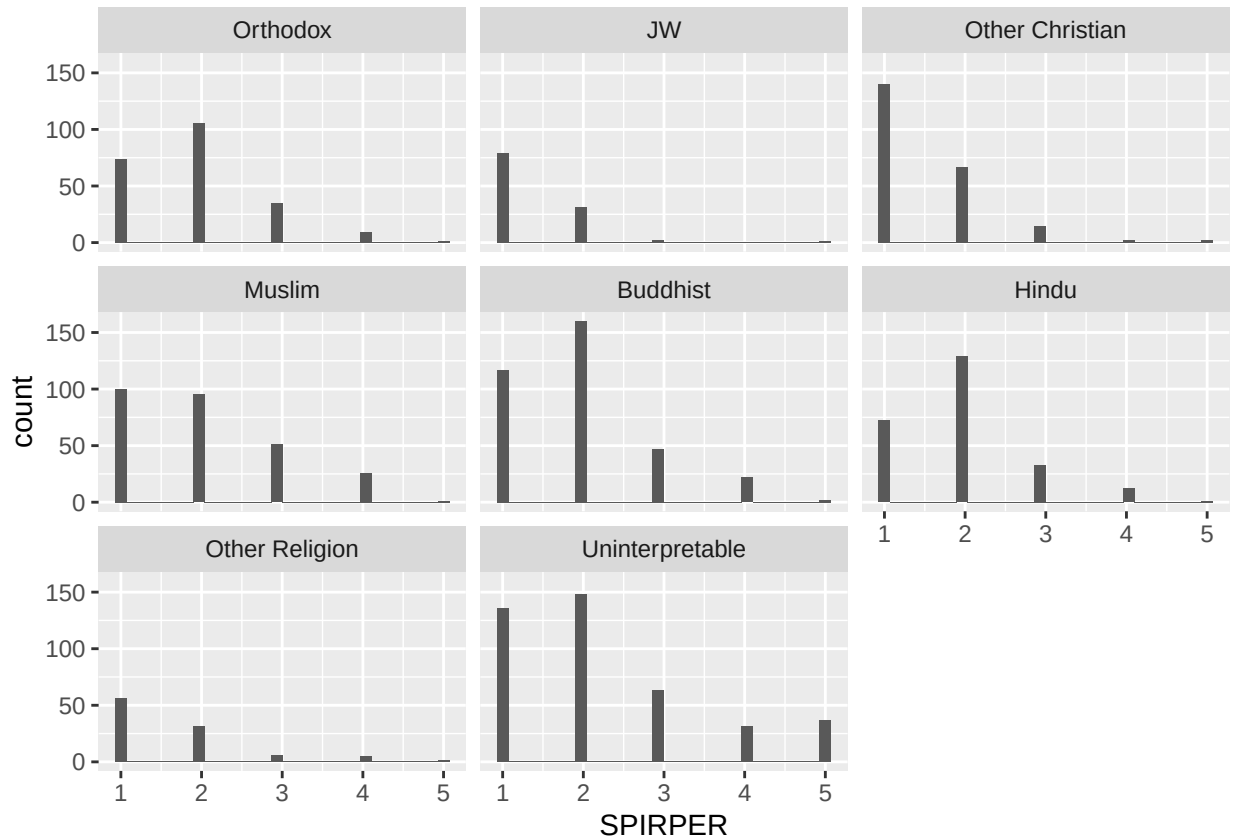
ggplot(data = data_subset1, aes(x=SPIRPER))+
  geom_histogram()+
  facet_wrap(~CURREL)
```



```
ggplot(data = data_subset2, aes(x=SPIRPER))+  
  geom_histogram()+  
  facet_wrap(~CURREL)
```



```
ggplot(data = data_subset3, aes(x=SPIRPER))+  
  geom_histogram()+  
  facet_wrap(~CURREL)
```

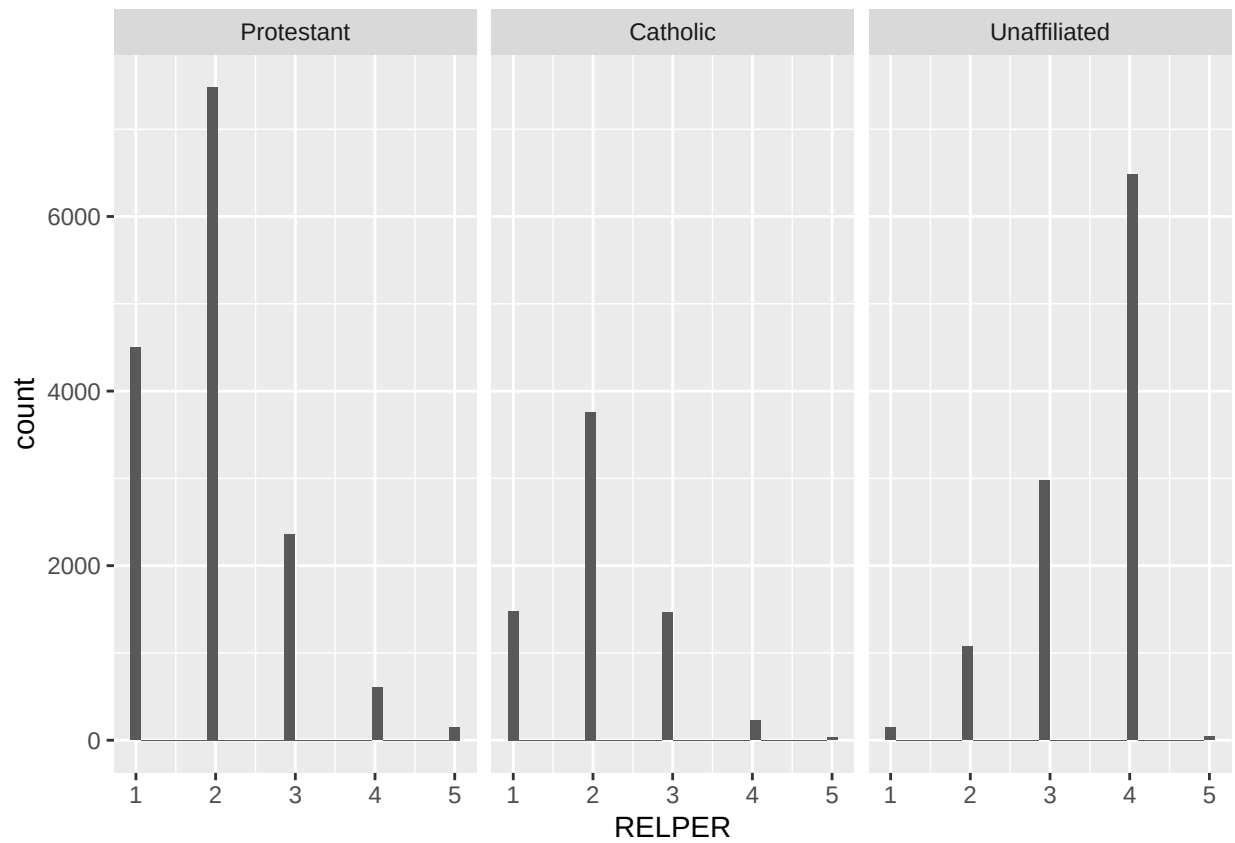


The spread for religiosity is shifted more to the higher values than spirituality: there were only two religions where the plurality (largest spike) was 1 (most religious).

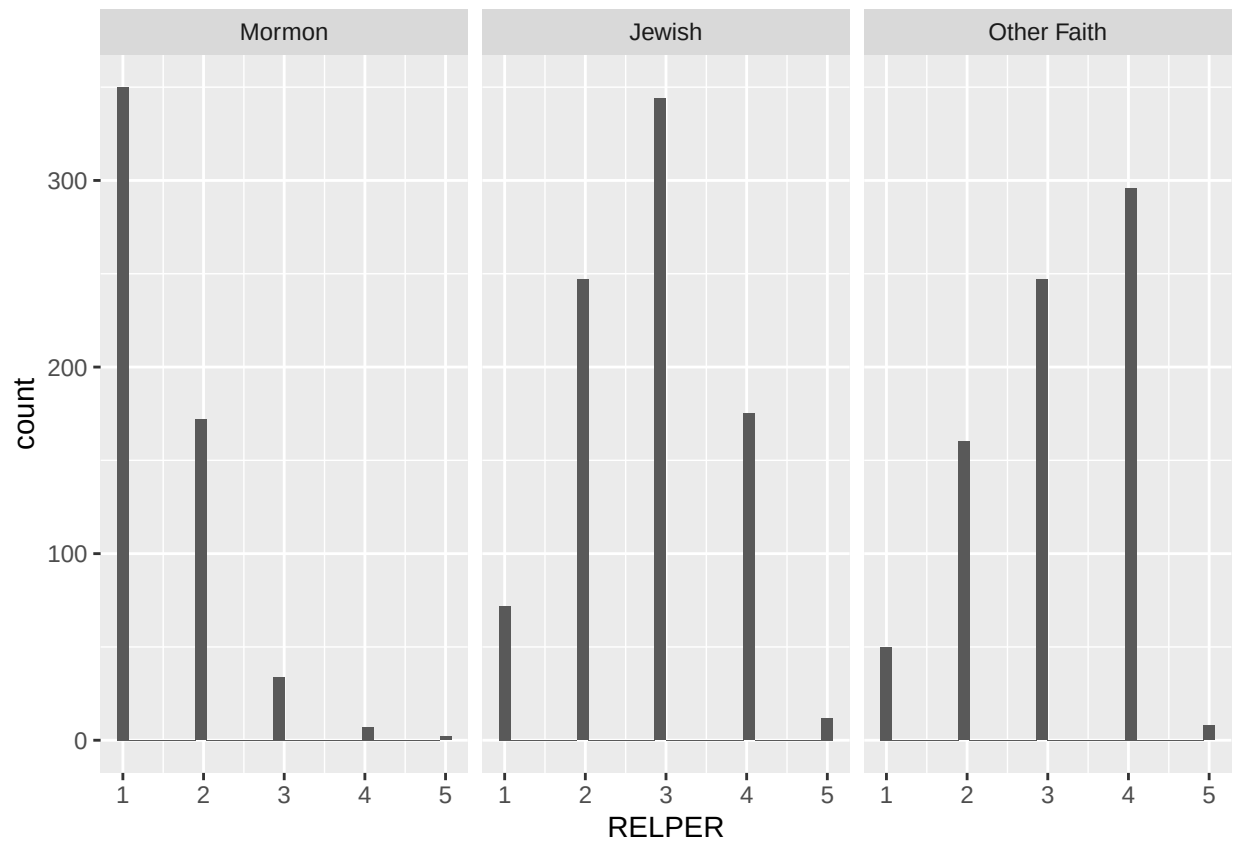
```
data$RELPER[data$RELPER == 99] <- 5

data_subset1 <- data |>
  filter(CURREL %in% c("Protestant", "Catholic", "Unaffiliated"))
data_subset2 <- data |>
  filter(CURREL %in% c("Mormon", "Jewish", "Other Faith"))
data_subset3 <- data |>
  filter(CURREL %in% c("Orthodox", "JW", "Other Christian", "Muslim", "Buddhist", "Hindu", "Other Relig

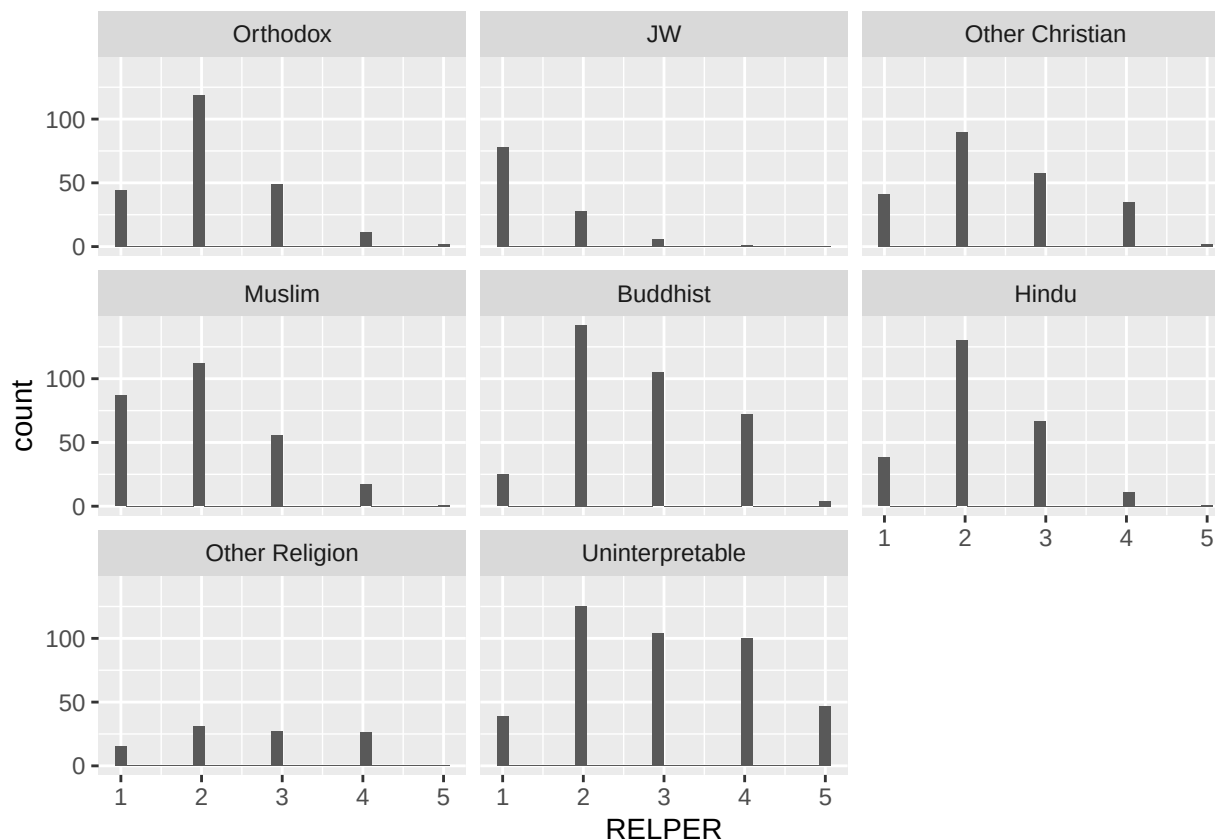
ggplot(data = data_subset1, aes(x=RELPER))+
  geom_histogram()+
  facet_wrap(~CURREL)
```



```
ggplot(data = data_subset2, aes(x=RELPER)) +  
  geom_histogram() +  
  facet_wrap(~CURREL)
```



```
ggplot(data = data_subset3, aes(x=RELPER))+  
  geom_histogram()+  
  facet_wrap(~CURREL)
```



Statistical Analysis

To formally investigate the relationships suggested by the exploratory data analysis, we apply several statistical methods, including analysis of variance (ANOVA), correlation analysis, and multiple linear regression. These methods allow us to test for group differences, measure associations between variables, and identify key predictors of religiosity.

Section 1: Group Differences

We begin by examining whether religiosity differs across religious traditions.

Hypothesis Test: H0: No difference in religiosity between groups. Ha: There is a difference.

```
data_filtered <- subset(data, RELPER != 5)

anova_model <- aov(RELPER ~ CURREL, data = data_filtered)
summary(anova_model)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## CURREL        13  17394  1338.0    2247 <2e-16 ***
## Residuals    36590   21785     0.6
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

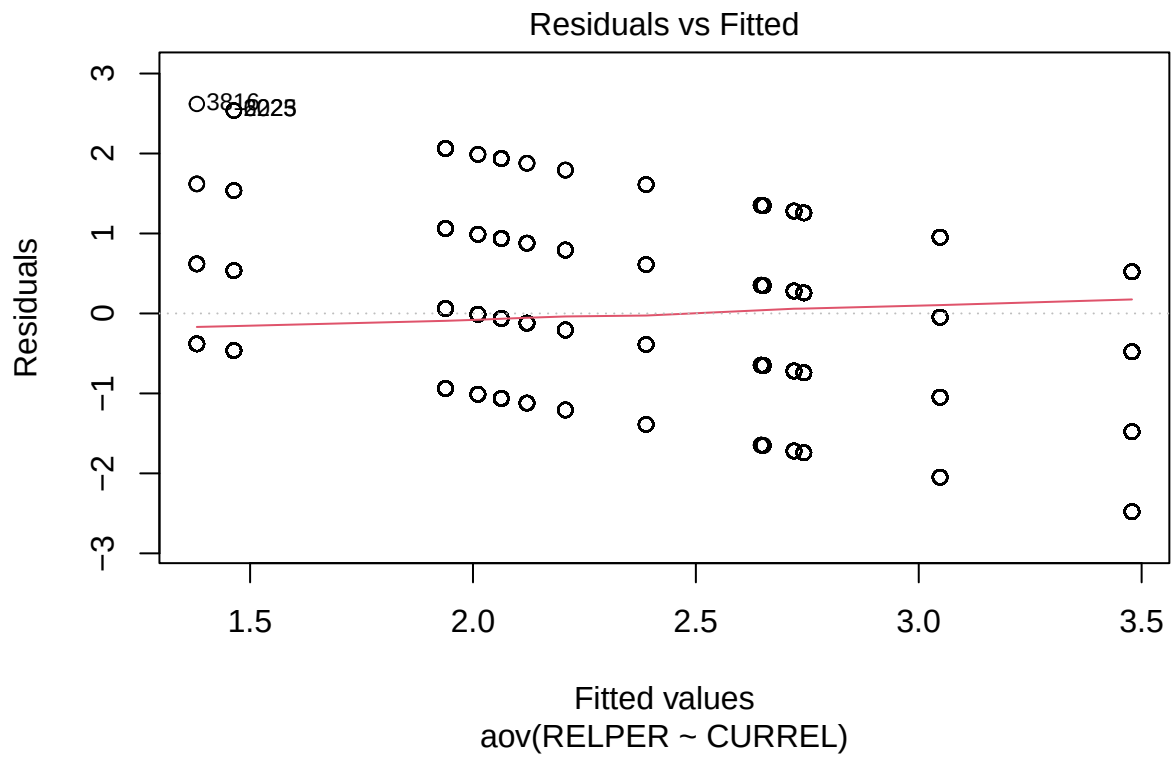
```
data_filtered %>%
  group_by(CURREL) %>%
  summarise(mean_religiosity = mean(RELPER, na.rm = TRUE),
            n = n()) %>%
  arrange(mean_religiosity)
```

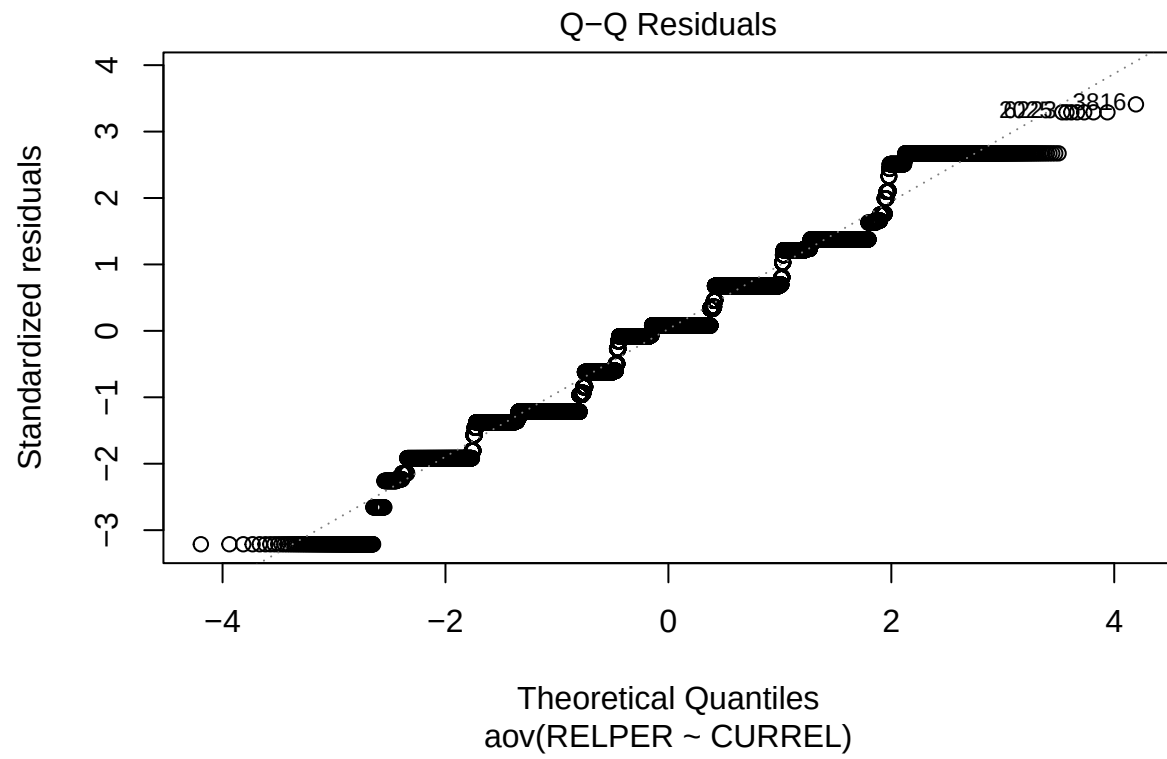
```
## # A tibble: 14 x 3
##   CURREL      mean_religiosity    n
##   <fct>          <dbl> <int>
## 1 JW              1.38   113
## 2 Mormon           1.46   563
## 3 Protestant       1.94 14949
## 4 Muslim           2.01   272
## 5 Catholic         2.06  6928
## 6 Orthodox         2.12   223
## 7 Hindu            2.21   246
## 8 Other Christian  2.39   224
## 9 Other Religion   2.65    99
## 10 Buddhist        2.65   344
## 11 Uninterpretable 2.72   368
## 12 Jewish           2.74   838
## 13 Other Faith     3.05   753
## 14 Unaffiliated    3.48 10684
```

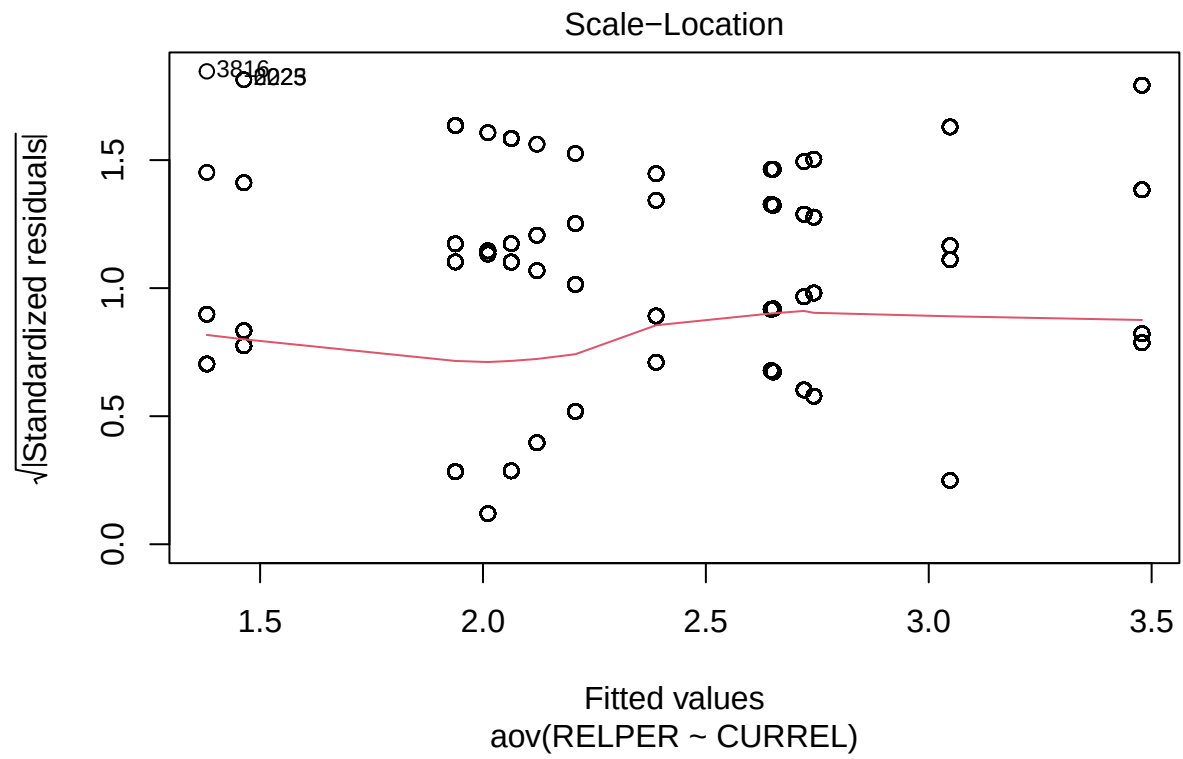
Interpret the what we get from anova model result...

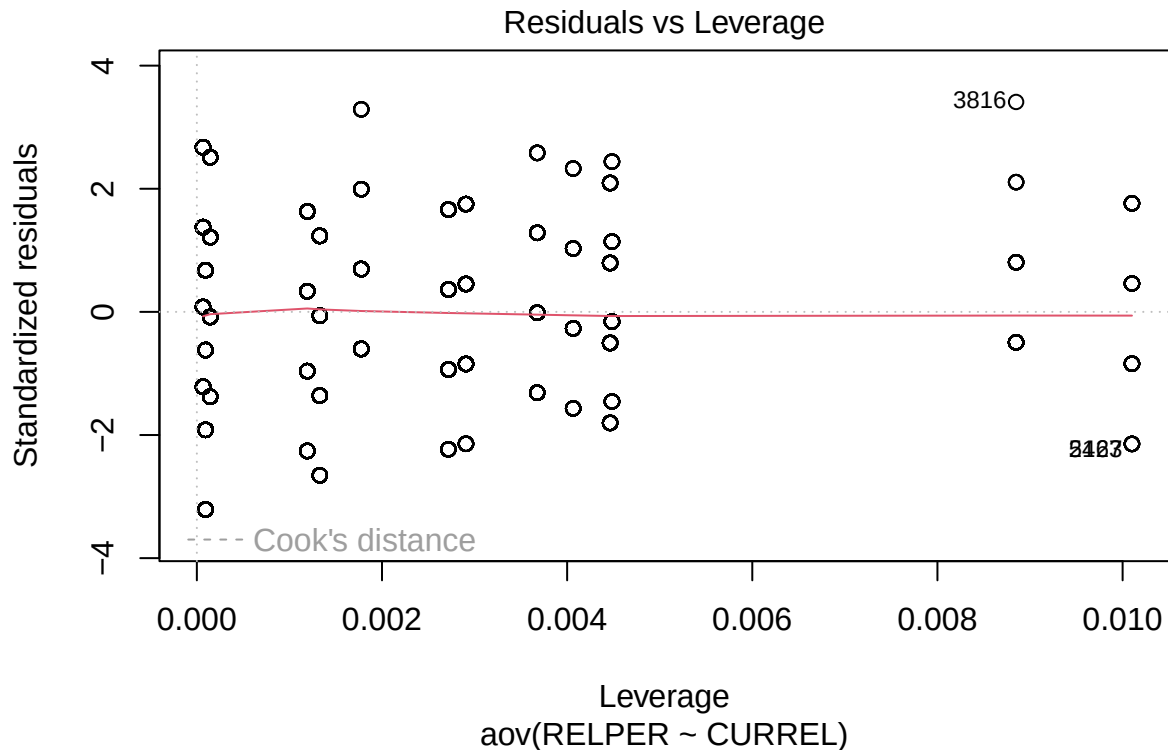
Check Assumptions for ANOVA: 1. Independence 2. Normality within groups 3. Equal variance

```
plot(anova_model)
```









... Interpret assumptions.

Section 2: Spirituality vs Religiosity

Next, we examine the relationship between spirituality and religiosity, as the exploratory analysis suggested that even unaffiliated individuals report relatively high levels of spirituality.

```
cor(data$RELPER, data$SPIRPER, use="complete.obs")
```

```
## [1] 0.5395335
```

```
model1 <- lm(RELPER ~ SPIRPER, data=data)
summary(model1)
```

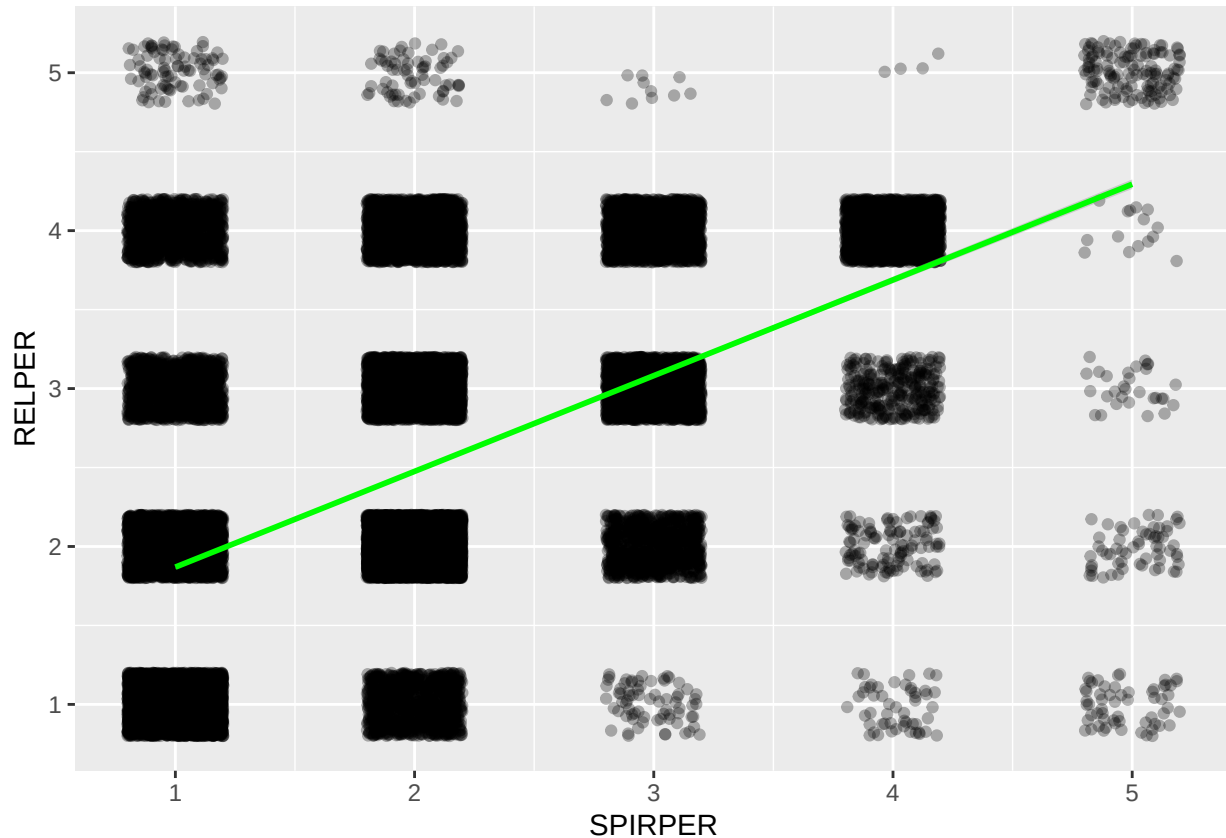
```
##
## Call:
## lm(formula = RELPER ~ SPIRPER, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2940 -0.4755 -0.0816  0.5245  3.1307
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 1.263160 0.010966 115.2 <2e-16 ***
## SPIRPER      0.606160 0.004924 123.1 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8887 on 36906 degrees of freedom
## Multiple R-squared:  0.2911, Adjusted R-squared:  0.2911
## F-statistic: 1.515e+04 on 1 and 36906 DF, p-value: < 2.2e-16
```

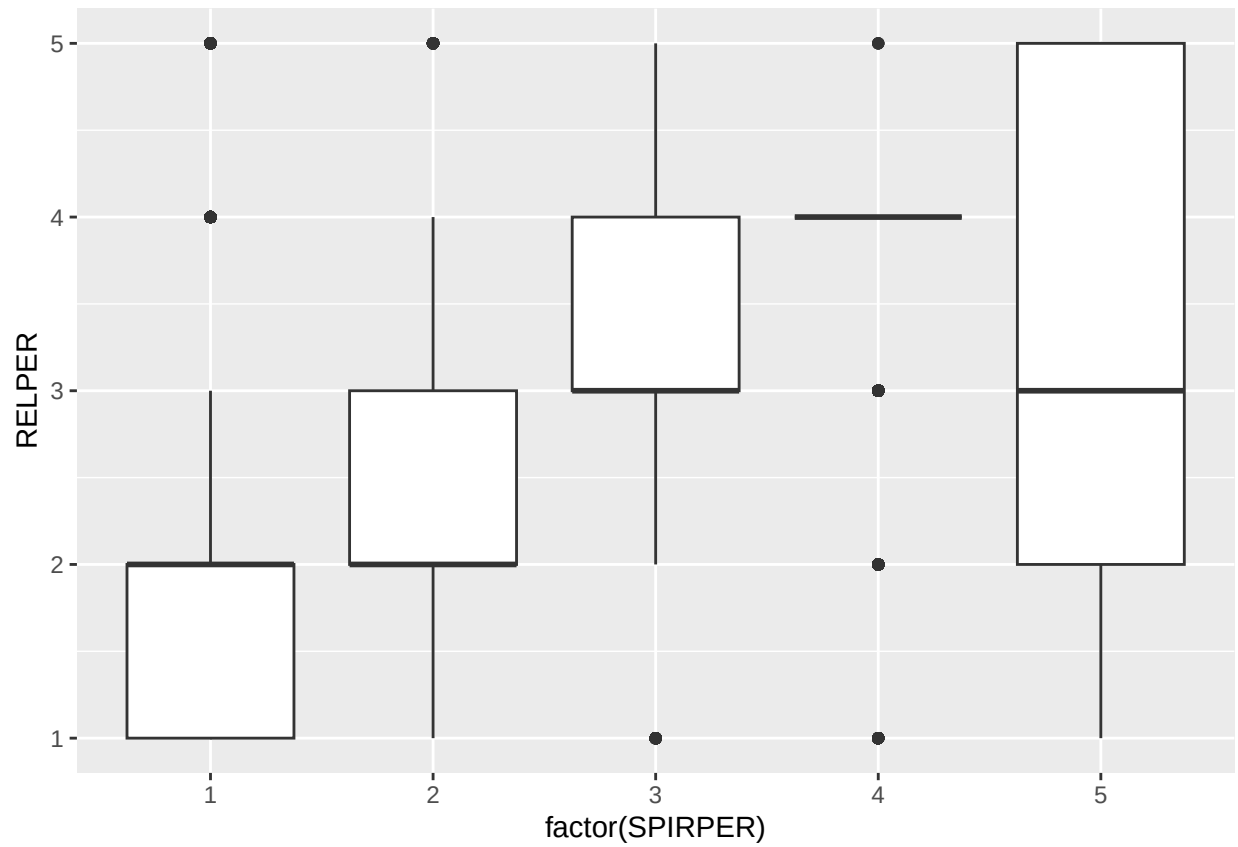
Choose a plot to see the relationship:

```
ggplot(data = data) +
  geom_jitter(aes(x = SPIRPER, y = RELPER), width = 0.2, height = 0.2, alpha = 0.3) +
  geom_smooth(method = "lm", aes(x = SPIRPER, y = RELPER), color = 'green')
```

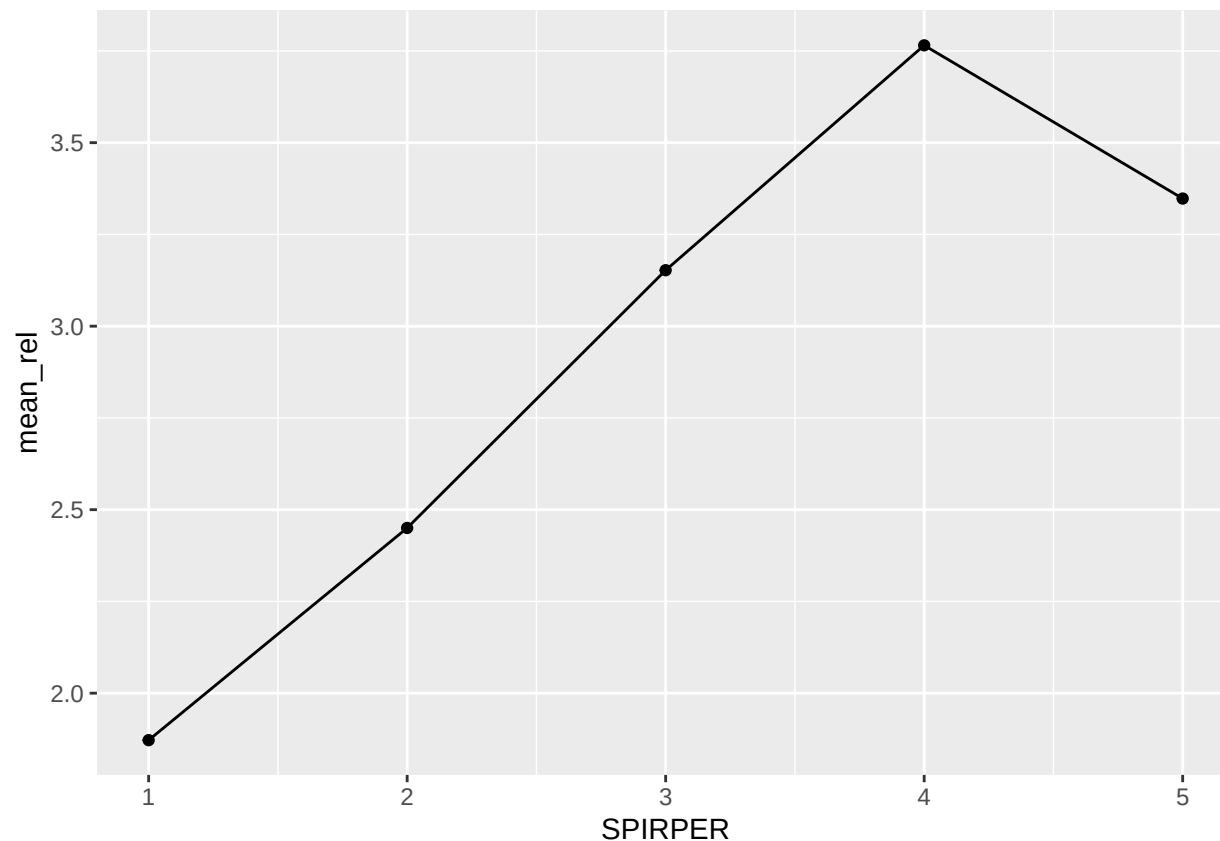
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
ggplot(data, aes(x = factor(SPIRPER), y = RELPER)) +
  geom_boxplot()
```



```
data |>  
  group_by(SPIRPER) |>  
  summarize(mean_rel = mean(RELPER, na.rm=TRUE)) |>  
  ggplot(aes(x = SPIRPER, y = mean_rel)) +  
  geom_point() +  
  geom_line()
```



Interpret...

Section 3: Behavioral Predictors

While spirituality captures internal beliefs, religiosity may also be influenced by observable behaviors such as attendance and prayer. We therefore examine whether these behavioral variables are significant predictors of religiosity.

```
no99 <- subset(data, RELPER != 5 & ATTNDPERRLS != 99 & PRAY != 99 & PRAC_B != 99 & PRAC_D != 99)
```

```
model2 <- lm(RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data=no99)
summary(model2)
```

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = no99)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.61517 -0.40885 -0.05211  0.40467  2.84929
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.705294   0.012384  56.953  <2e-16 ***
## ATTNDPERRLS  0.215591   0.003086  69.868  <2e-16 ***
```

```
## PRAY          0.186483    0.002190   85.169   <2e-16 ***
## PRAC_B        0.063182    0.003543   17.833   <2e-16 ***
## PRAC_D       -0.004961    0.003687   -1.345    0.178
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6659 on 35842 degrees of freedom
## Multiple R-squared:  0.5856, Adjusted R-squared:  0.5855
## F-statistic: 1.266e+04 on 4 and 35842 DF,  p-value: < 2.2e-16
```

```
protestant_data <- subset(no99, CURREL == "Protestant")
model_protestant <- lm(RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D,
                      data = protestant_data)
summary(model_protestant)
```

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = protestant_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.8208 -0.4932 -0.0820  0.3761  2.6364
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.054964   0.016932  62.307 < 2e-16 ***
## ATTNDPERRLS  0.119634   0.004542  26.341 < 2e-16 ***
## PRAY         0.102805   0.004215  24.391 < 2e-16 ***
## PRAC_B       0.067325   0.005328  12.637 < 2e-16 ***
## PRAC_D       0.018922   0.005331   3.549 0.000387 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.669 on 14560 degrees of freedom
## Multiple R-squared:  0.2734, Adjusted R-squared:  0.2732
## F-statistic: 1370 on 4 and 14560 DF,  p-value: < 2.2e-16
```

```
catholic_data <- subset(no99, CURREL == "Catholic")
model_catholic <- lm(RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D,
                    data = catholic_data)
summary(model_catholic)
```

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = catholic_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1253 -0.3578 -0.0318  0.3484  2.8866
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.679348   0.025274  26.879 < 2e-16 ***
```

```
## ATTNDPERRLS 0.174389 0.005501 31.700 < 2e-16 ***
## PRAY 0.129122 0.004237 30.475 < 2e-16 ***
## PRAC_B 0.067755 0.006135 11.044 < 2e-16 ***
## PRAC_D 0.031407 0.007179 4.375 1.23e-05 ***
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5383 on 6809 degrees of freedom
## Multiple R-squared: 0.4764, Adjusted R-squared: 0.4761
## F-statistic: 1549 on 4 and 6809 DF, p-value: < 2.2e-16
```

```
muslim_data <- subset(no99, CURREL == "Muslim")
model_muslim <- lm(RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D,
  data = muslim_data)
summary(model_muslim)
```

```
##
## Call:
## lm(formula = RELPER ~ ATTNDPERRLS + PRAY + PRAC_B + PRAC_D, data = muslim_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.90472 -0.51176 -0.07219  0.41528  2.37060
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.85483    0.13110   6.521 3.54e-10 ***
## ATTNDPERRLS  0.13021    0.02980   4.369 1.79e-05 ***
## PRAY         0.16148    0.02480   6.512 3.72e-10 ***
## PRAC_B       0.06696    0.03438   1.948  0.0525 .
## PRAC_D       0.02527    0.03519   0.718  0.4732
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6871 on 264 degrees of freedom
## Multiple R-squared: 0.4063, Adjusted R-squared: 0.3973
## F-statistic: 45.16 on 4 and 264 DF, p-value: < 2.2e-16
```

Hypothesis Testing: H0: Ha:

Interpret the result...

Check for Assumptions of Regression: 1. Linearity 2. Nearly normal residuals 3. Constant variability

Section 4: Political Predictors of Religiosity

```
data_clean <- subset(data, RELPER != 99 & ABRTLGL != 99 & GAYMARR != 99 & QB2A != 99 & CHNG_C)
model3 <- lm(RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = data_clean)
summary(model3)
```

```
##
```

```
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.1085 -0.6519 -0.1085  0.8915  3.7058
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.6830172  0.0156302 235.634 <2e-16 ***
## ABRTLGL      -0.2550419  0.0065717 -38.809 <2e-16 ***
## GAYMARR      -0.3135173  0.0077475 -40.467 <2e-16 ***
## QB2A         -0.0049037  0.0166560  -0.294  0.7684
## CHNG_C       -0.0010580  0.0004984  -2.123  0.0338 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.921 on 35221 degrees of freedom
## Multiple R-squared:  0.2378, Adjusted R-squared:  0.2377
## F-statistic: 2748 on 4 and 35221 DF, p-value: < 2.2e-16

data_clean_spir <- subset(data, SPIRPER != 99 & ABRTLGL != 99 & GAYMARR != 99 & QB2A != 99 & CHNG_C)

data_spir <- lm(SPIRPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = data_clean)
summary(data_spir)

##
## Call:
## lm(formula = SPIRPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = data_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3393 -0.6044 -0.1779  0.6618  3.5559
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.683e+00  1.527e-02 175.670 < 2e-16 ***
## ABRTLGL      -1.603e-01  6.420e-03 -24.966 < 2e-16 ***
## GAYMARR      -1.145e-01  7.569e-03 -15.133 < 2e-16 ***
## QB2A         -6.954e-02  1.627e-02  -4.274 1.93e-05 ***
## CHNG_C       1.156e-05  4.870e-04   0.024  0.981
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.8998 on 35221 degrees of freedom
## Multiple R-squared:  0.08104, Adjusted R-squared:  0.08094
## F-statistic: 776.5 on 4 and 35221 DF, p-value: < 2.2e-16
```

It would appear that removal of the 99s which represent those who did not answer made a substantial difference on the fit of the model. Furthermore, support for gay marriage predicts religiosity well, as does support for transgender individuals. However, acceptance of homosexuality is an incredibly poor predictor of religiosity in the body public.

Alternatively, support for abortion, homosexuality, and gay marriage predicts *spirituality* but support for transgender individuals does not predict *spirituality*

```
data_clean2 <- subset(data, RELPER != 99 & ABRTLGL != 99 & GAYMARR != 99 & QB2A != 99 & CHNG_C != 99)
```

```
catholic_pol <- data_clean2 |> filter(CURREL == "Catholic")
catholic_polly <- lm(RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = catholic_pol)
summary(catholic_polly)
```

```
##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = catholic_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.5059 -0.4432 -0.1394  0.4941  3.5568
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.843525   0.032614  87.186 <2e-16 ***
## ABRTLGL      -0.225040   0.011091 -20.290 <2e-16 ***
## GAYMARR      -0.137512   0.013897  -9.895 <2e-16 ***
## QB2A         0.026926   0.027925   0.964  0.335
## CHNG_C       -0.001988   0.011860  -0.168  0.867
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7036 on 6512 degrees of freedom
## Multiple R-squared:  0.1519, Adjusted R-squared:  0.1514
## F-statistic: 291.6 on 4 and 6512 DF,  p-value: < 2.2e-16
```

```
protestant_pol <- data_clean2 |> filter(CURREL == "Protestant")
protestant_polly <- lm(RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = protestant_pol)
summary(protestant_polly)
```

```
##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = protestant_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3114 -0.6037 -0.0915  0.3070  3.3963
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.579250   0.026161  98.591 < 2e-16 ***
## ABRTLGL      -0.089316   0.008836 -10.108 < 2e-16 ***
## GAYMARR      -0.124174   0.010350 -11.998 < 2e-16 ***
## QB2A         -0.064033   0.020881  -3.067  0.00217 **
## CHNG_C        0.003236   0.010320   0.314  0.75384
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7991 on 14034 degrees of freedom
```

```
## Multiple R-squared:  0.07049,    Adjusted R-squared:  0.07023
## F-statistic: 266.1 on 4 and 14034 DF,  p-value: < 2.2e-16
```

```
unaffiliated_pol <- data_clean2 |> filter(CURREL == "Unaffiliated")
unaffiliated_polly <- lm(RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = unaffiliated_pol)
summary(unaffiliated_polly)
```

```
##
## Call:
## lm(formula = RELPER ~ ABRTLGL + GAYMARR + QB2A + CHNG_C, data = unaffiliated_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.7160 -0.5340  0.2840  0.4403  2.2287
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.159420   0.029526 140.874  <2e-16 ***
## ABRTLGL      -0.156293   0.010387 -15.046  <2e-16 ***
## GAYMARR      -0.172493   0.013189 -13.079  <2e-16 ***
## QB2A         -0.023650   0.030521  -0.775    0.438
## CHNG_C       -0.091006   0.009786  -9.300  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6885 on 10420 degrees of freedom
## Multiple R-squared:  0.1083, Adjusted R-squared:  0.108
## F-statistic: 316.4 on 4 and 10420 DF,  p-value: < 2.2e-16
```

While support for abortion, same sex marriage, and homosexuality is a statistically significant predictor of religiosity in Catholics and people who are unaffiliated, support for these causes is not statistically significant in predicting religiosity in Protestants. Interestingly, greater acceptance of transgender individuals cannot be used to predict religiosity in people who are unaffiliated.

```
data_clean3 <- subset(data, RELPER != 99 & CHNG_A != 99 & CHNG_B != 99 & DIVRELPOP != 99 & DIVRACPOP !=
catholic_pol2 <- data_clean3 |> filter(CURREL == "Catholic")
catholic_polly2 <- lm(RELPER ~ CHNG_A + CHNG_B + DIVRELPOP + DIVRACPOP + QB2C, data = catholic_pol)
summary(catholic_polly2)
```

```
##
## Call:
## lm(formula = RELPER ~ CHNG_A + CHNG_B + DIVRELPOP + DIVRACPOP +
##      QB2C, data = catholic_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.28838 -0.09125 -0.08980  0.90463  3.14626
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.0916628  0.0108237 193.249  <2e-16 ***
```

```
## CHNG_A      0.0007251  0.0009820  0.738  0.4603
## CHNG_B     -0.0020007  0.0015930 -1.256  0.2092
## DIVRELPOP   -0.0023070  0.0013777 -1.675  0.0941 .
## DIVRACPOP    0.0020643  0.0015143  1.363  0.1729
## QB2C        -0.0001745  0.0008096 -0.216  0.8294
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7637 on 6511 degrees of freedom
## Multiple R-squared:  0.000831, Adjusted R-squared:  6.374e-05
## F-statistic: 1.083 on 5 and 6511 DF, p-value: 0.3674
```

```
data_clean3 <- subset(data, RELPER != 99 & CHNG_A != 99 & CHNG_B != 99 & DIVRELPOP != 99 & DIVRACPOP !=
protestant_pol2 <- data_clean3 |> filter(CURREL == "Protestant")
protestant_polly2 <- lm(RELPER ~ CHNG_A + CHNG_B + DIVRELPOP + DIVRACPOP + QB2C, data = catholic_pol)
summary(protestant_polly2)
```

```
##
## Call:
## lm(formula = RELPER ~ CHNG_A + CHNG_B + DIVRELPOP + DIVRACPOP +
##      QB2C, data = catholic_pol)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.28838 -0.09125 -0.08980  0.90463  3.14626
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.0916628  0.0108237 193.249  <2e-16 ***
## CHNG_A       0.0007251  0.0009820   0.738  0.4603
## CHNG_B     -0.0020007  0.0015930  -1.256  0.2092
## DIVRELPOP   -0.0023070  0.0013777  -1.675  0.0941 .
## DIVRACPOP    0.0020643  0.0015143   1.363  0.1729
## QB2C        -0.0001745  0.0008096  -0.216  0.8294
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7637 on 6511 degrees of freedom
## Multiple R-squared:  0.000831, Adjusted R-squared:  6.374e-05
## F-statistic: 1.083 on 5 and 6511 DF, p-value: 0.3674
```

Support for a growing immigrant population is not predictive of religiosity, but

Full Model, Search for the Best Model

Finally, we construct a comprehensive model that incorporates spirituality, behavioral measures, belief variables, and religious affiliation to determine which factors are most important in predicting religiosity.

```
final_model <- lm(RELPER ~ SPIRPER + ATTNDPERRLS + PRAY + PRAC_B + PRAC_D + GOD + HVN + HLL + CURREL, d
summary(final_model)
```

```
##
## Call:
## lm(formula = RELPER ~ SPIRPER + ATTNDPERRLS + PRAY + PRAC_B +
##      PRAC_D + GOD + HVN + HLL + CURREL, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.0495 -0.6245 -0.0636  0.3609  3.6930
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.2678354   0.0098796  128.329 < 2e-16 ***
## SPIRPER         0.3752479   0.0044903   83.568 < 2e-16 ***
## ATTNDPERRLS     0.0090659   0.0005188   17.475 < 2e-16 ***
## PRAY            0.0055012   0.0004691   11.726 < 2e-16 ***
## PRAC_B          0.0020458   0.0004775    4.284 1.84e-05 ***
## PRAC_D        -0.0006579   0.0005129   -1.283 0.199628
## GOD             0.0006985   0.0003299    2.117 0.034239 *
## HVN             0.0007836   0.0002839    2.760 0.005781 **
## HLL            -0.0004881   0.0002564   -1.904 0.056983 .
## CURELCatholic   0.0285637   0.0106052    2.693 0.007077 **
## CURELMormon    -0.3601116   0.0312536  -11.522 < 2e-16 ***
## CURELOrthodox   0.0900067   0.0489390    1.839 0.065900 .
## CURELJW        -0.4304608   0.0688268   -6.254 4.04e-10 ***
## CURELOther Christian 0.5166936   0.0488330   10.581 < 2e-16 ***
## CURELJewish     0.5405888   0.0258636   20.902 < 2e-16 ***
## CURELMuslim    -0.0588955   0.0445109   -1.323 0.185787
## CURELBuddhist   0.5906092   0.0395321   14.940 < 2e-16 ***
## CURELHindu      0.1577924   0.0467441    3.376 0.000737 ***
## CURELOther Religion 0.6953460   0.0734584    9.466 < 2e-16 ***
## CURELOther Faith 1.1236508   0.0271105   41.447 < 2e-16 ***
## CURELUnaffiliated 1.1451878   0.0101093  113.280 < 2e-16 ***
## CURELUninterpretable 0.6820045   0.0367126   18.577 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7284 on 36886 degrees of freedom
## Multiple R-squared:  0.524, Adjusted R-squared:  0.5237
## F-statistic: 1934 on 21 and 36886 DF, p-value: < 2.2e-16
```

Check for Assumptions of Regression: 1. Linearity 2. Nearly normal residuals 3. Constant variability

Dropping the variables that has p-value higher than 0.05. After dropping will model be better?